From pointwise Ricci control to finite monodromy and exact realization

24 September 2026. This is a composition theorem for a specified geometric class, not a theorem that arbitrary near-Einstein four-manifolds admit a surface fibration. Its global input is the thick-fiber cycle theorem in the companion note. Its curvature identity is classical and was already used in Round 20, Sections 5–7. The new project-level conclusion is a quantitative **sufficient condition for exact existence**, including zero-signature bundles: small pointwise Ricci error detects finite mapping-class monodromy, instead of assuming it.

1. Statement

Let

$$\Sigma_f \longrightarrow M \xrightarrow{\pi} B$$

be a smooth oriented surface bundle, with fiber genus $f \geqslant 2$ and closed oriented base of genus at least two. A Riemannian metric g makes $\pi : (M, g) \to (B, b)$ a Riemannian submersion. Write h_y for its vertical metric, $\mathscr{H}$ for its orthogonal horizontal distribution, and K_b for the Gaussian curvature of the base. Assume the fibers are minimal. They need not be totally geodesic, intrinsically hyperbolic, or mutually isometric. The horizontal distribution need not be integrable.

Fix a reference fiber over y_0. Uniformize its conformal class to curvature -1, and assume its hyperbolic systole is at least $\iota > 0$. Define

$$\begin{aligned} \delta &= \| \operatorname{Ric}_g^\sharp + I \|_{L^\infty, \mathrm{op}}, \\ \eta &= \| (K_b + 1)_+ \|_{L^\infty(B)}, \qquad D_B = \operatorname{diam}(B, b). \end{aligned} \tag{1}$$

The positive part in η is intentional; no two-sided curvature closeness of the base is needed for the estimate below.

Theorem 1 (Ricci-to-monodromy realization). There is $\epsilon_{f,\iota} > 0$, depending only on fiber genus and the reference systole, such that

$$2D_B \sqrt{\delta + \eta} < \epsilon_{f,\iota} \tag{2}$$

implies that the mapping-class monodromy of the bundle has finite image. Consequently the **same oriented smooth manifold** M carries a negative Kähler–Einstein metric $\widehat{g}$, with

$$\operatorname{Ric}_{\widehat{g}} = -\widehat{g}, \qquad (\widetilde{M}, \tilde{\widehat{g}}) \cong \mathbb{H}^2 \times \mathbb{H}^2. \tag{3}$$

The theorem imposes no bound on total-space diameter, total volume, injectivity radius, full curvature, or a spectral inverse. It does not assert closeness of $\widehat{g}$ to g.

In particular, when $K_b = -1$, condition (2) is simply

$$\delta < \frac{\epsilon_{f,\iota}^2}{4 D_B^2}. \tag{4}$$

Thus, within this explicitly specified input class, the output is already the Einstein existence requested by the negative side of Yau26. The additional fibration, minimality, thickness and base-scale hypotheses are the restriction; obtaining KE rather than a general Einstein metric is not a defect of the existence conclusion.

2. Horizontal Ricci controls conformal deformation

For a horizontal vector X, define the symmetric vertical endomorphism S_X by

$$h(S_X U, V) = \langle (\nabla_U V)^{\mathscr{H}}, X \rangle.$$

Let $A_X Y = \frac{1}{2}[X, Y]^{\mathscr{V}}$ for basic horizontal vector fields. Minimality is exactly $\operatorname{tr} S_X = 0$ for every horizontal X. For a unit horizontal vector, O'Neill's contracted curvature formulas give

$$\mathrm{Ric}_g(X, X) = K_b - |S_X|^2_{\mathrm{HS}} - 2\sum_a |A_X e_a|^2. \tag{5}$$

Here e_a is a horizontal orthonormal basis. The derivative of the fiber mean-curvature vector in the general formula is zero because that vector is identically zero, not because its integral vanishes. To check the coefficients, horizontal sectional curvatures contribute $K_b - 3\sum_a |A_X e_a|^2$; the mixed sectional curvatures contribute $\sum_a |A_X e_a|^2 - |S_X|^2_{\mathrm{HS}}$. This is the same identity used in Round 20; our sign choice for S has no effect on (5). The primary source is O'Neill, *The fundamental equations of a submersion.*

Since the horizontal compression of $\mathrm{Ric}_g + g$ has operator norm at most δ, (5) yields, pointwise,

$$\begin{aligned} |S_X|^2_{\mathrm{HS}} + 2\sum_a |A_X e_a|^2 &= (K_b + 1) - (\mathrm{Ric}_g + g)(X, X) \\ &\leqslant \eta + \delta. \end{aligned} \tag{6}$$

In particular, for an arbitrary horizontal vector,

$$|S_X|_{\mathrm{op}} \leqslant \sqrt{\delta + \eta}\,|X|. \tag{7}$$

No vertical or mixed Ricci equation is discarded in the hypothesis: the full Ricci defect in (1) implies the horizontal bound. Conversely we make no claim that the horizontal bound alone makes an arbitrary trial metric near-Einstein. Intrinsic fiber curvature is absent from (5), which is why no fiber uniformization hypothesis is required away from y_0.

3. Integrating the actual deformation around a loop

Let $\gamma : [0, 1] \to B$ be a piecewise smooth loop based at y_0. Its horizontal transport $P_t : F_{y_0} \to F_{\gamma(t)}$ is defined for the entire interval because the bundle is compact. Set $k_t = P_t^* h_{\gamma(t)}$. Differentiating on the fixed marked fiber,

$$k_t^{-1}\dot{k}_t = -2P_t^* S_{\dot{\gamma}^{\mathscr{H}}}. \tag{8}$$

The trace is zero by minimality. Write x_t for the corresponding point in Teichmüller space, with metric normalized by $d_T = \frac{1}{2}\inf \log K$.

For clarity the deformation estimate does not require differentiating the uniformization map. For any nonzero tangent vector on the reference fiber,

$$\left| \frac{d}{dt} \log |v|_{k_t} \right| \leqslant \sup_{F_{\gamma(t)}} |S_{\dot{\gamma}^{\mathscr{H}}}|_{\mathrm{op}}.$$

After integration, the ratio between the largest and smallest singular stretch of the identity map is at most $\exp(2\int \sup |S|\, dt)$. This ratio is its quasiconformal dilatation. It follows that

$$\begin{aligned} d_T(x_0, x_1) &\leqslant \int_0^1 \sup_{F_{\gamma(t)}} |S_{\dot{\gamma}^{\mathscr{H}}}|_{\mathrm{op}}\, dt \\ &\leqslant \sqrt{\delta + \eta}\ \mathrm{Length}_b(\gamma). \end{aligned} \tag{9}$$

The endpoint is the action on x_0 of the monodromy class, or its inverse according to the marking convention. Distances are the same. Connection holonomy itself need not be a homomorphism into diffeomorphisms when the connection is curved; its **mapping-class** holonomy is the topological monodromy homomorphism. Formula (9) controls that homomorphism without falsely imposing flatness of the connection.

4. Short generators and a common finite group

We use the following elementary global fact.

Lemma 2. For a closed connected Riemannian manifold B, any basepoint y_0, and $a > 0$, its fundamental group has a finite generating set represented by loops of length less than $2\,\mathrm{diam}(B) + a$.

Proof. Subdivide any based loop into arcs of length less than a. Join each subdivision vertex to y_0 by a minimizing path of length at most D_B. The original loop is the product of the resulting triangular based loops; cancellation is literal along the joining paths. Each triangular loop has length less than $2D_B + a$. Only finitely many based homotopy classes admit a representative with that length bound: lifting to the complete universal cover associates each class to a distinct deck translate of a fixed lift of y_0 in a compact ball. The deck orbit is discrete and the action is proper. Thus the collection of all these short classes is a finite generating set. □

The companion theorem supplies the following precise Teichmüller fact: for every f, ι, there exists $\epsilon_{f,\iota} > 0$ such that, at every marked hyperbolic surface of systole at least ι, the subgroup generated by **all** mapping classes moving it less than $\epsilon_{f,\iota}$ is finite. This is stronger than saying that each such class separately has finite order. It follows from proper discontinuity of the mapping-class action and compactness of the thick moduli quotient, using neighborhoods of a **common** finite stabilizer. The companion note proves the uniform statement and treats the possible ineffective finite kernel in genus two.

By the strict inequality (2), choose $a > 0$ sufficiently small that

$$(2D_B + a)\sqrt{\delta + \eta} < \epsilon_{f,\iota}.$$

Apply (9) to the finite generating set of Lemma 2. All its monodromy elements lie in the same finite subgroup supplied by the thick-point lemma. Therefore the entire monodromy image is finite. This is the global step that a pointwise estimate for S alone does not provide.

5. Realization on the original smooth bundle

Let G be the finite monodromy image. Nielsen realization gives a hyperbolic metric h_F on the model fiber and an action $G \hookrightarrow \mathrm{Isom}^+(\Sigma_f, h_F)$ realizing those mapping classes. Let $\widehat{B} \to B$ be the connected regular cover corresponding to the kernel of monodromy. Its deck group is G. Uniformize any conformal structure on the oriented base to a hyperbolic metric $\widehat{b}$ and lift it to $\widehat{B}$. Then

$$(\widehat{B} \times \Sigma_f)/G \tag{10}$$

is a smooth diagonal quotient; the action is free because it is free on the base factor. It carries the quotient of the product metric, with Ricci tensor equal to minus the metric, and the quotient complex structure is Kähler.

The identity component of the orientation-preserving diffeomorphism group of a closed surface of genus at least two is contractible. Thus oriented smooth surface bundles are classified by their

mapping-class monodromy; there is no additional identity-component twisting over B. The quotient bundle (10) and the original bundle are smoothly isomorphic, since their monodromy representations coincide. Pulling back the quotient metric proves (3) on the original M. The companion note gives the classical Nielsen and surface-diffeomorphism references. Neither the original horizontal distribution nor its full diffeomorphism holonomy has to agree with the resulting product quotient.

6. A quantified escape alternative

Taking the contrapositive gives a useful theorem about actual inputs, not an assertion of general Einstein nonexistence.

Corollary 3. If the mapping-class monodromy image is infinite and the reference fiber is ι-thick, then every minimal-submersion metric as above satisfies

$$(\delta + \eta) D_B^2 \geqslant \frac{\epsilon_{f,\iota}^2}{4}. \tag{11}$$

For hyperbolic bases and $\delta_j \to 0$, a sequence of such metrics with fixed fiber genus and a uniformly thick reference fiber therefore has

$$D_{B,j} \geqslant \frac{\epsilon_{f,\iota}}{2\sqrt{\delta_j}} \longrightarrow \infty. \tag{12}$$

Since a Riemannian submersion is distance nonincreasing, $\operatorname{diam}(M, g) \geqslant D_B$. Alternatively, with bounded D_B, infinite monodromy and vanishing defect force the reference conformal systole to approach zero along any subsequence for which it could otherwise have a positive lower bound.

These alternatives distinguish two real escape mechanisms: long horizontal transport and thin fiber conformal geometry. The companion note supplies Dehn-twist examples separating them. They do not say that the same smooth manifold has no Einstein metric outside this submersion class, nor that every near-Einstein manifold has either of these two geometric descriptions.

7. What changes, and what does not

Round 20 already proved a signature-energy obstruction for minimal hyperbolic-fiber submersions and recovered the classical exact adapted Einstein classification. For a signature-zero bundle its signature lower bound vanishes. The present argument instead uses the actual mapping-class action and one thick conformal fiber. It can detect infinite monodromy even when signature is zero; more importantly, it proves existence when the smallness threshold is met.

This is not a new construction of bidisk quotients: those are classical. It is a geometric criterion that **recognizes** their smooth bundle types from the input metric, with finite monodromy deduced rather than assumed. No assertion of worldwide priority is made. The first-line connection is that local or average Kähler-type structure requires global transport information to produce existence. Here that information is obtained from a definite horizontal Ricci identity and a thick-point monodromy threshold, rather than from another local calibration coefficient.

The remaining step toward arbitrary Yau26 inputs is substantial: neither an actual surface submersion nor fiber minimality nor thickness is supplied by the general pointwise Ricci hypothesis. The existing collapsed-bulk theorem does not automatically supply them across its exceptional set. This theorem completes the existence argument for the stated input class without claiming that missing classification.